Virtual Environment for the Evaluation of Gestures in Odontology

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Abstract: This work presents VEEGO, a Virtual Environment (VE) for the automatic Evaluation of Gestures in Odontology and its pipeline for its use. The gestures of learners and teachers are recorded with motion capture while performing a care protocol in a real preclinical environment. Their performances are replayed, through their 3D avatars in VE, and compared in terms of motor aspects, named "Observation Needs" (ON) which are defined by teachers. The evaluation relies on a new approach based on: (i) configurable descriptors made of directional vectors and orientations for each ON and (ii), automatically generated Random Forests (RF), one per ON, linking descriptor values to each ON. An experiment was conducted in which teachers' motions were captured on 5 different tasks to demonstrate 5 ONs shared among them *i.e.* "improper positioning around the patient", "overly curved back", "back leaning to the side", "raised elbow" and "no fulcrum". The trained RFs showed a very high F1 scores on unseen gestures. The proposed VEEGO pipeline is designed to: (i) aggregate the motor expertise of multiple experts free to express and demonstrate their motor expertise, (ii) indicate to students the motor aspects to improve through colours of their 3D avatar body parts and textual indicators and (iii) consider the evolution of ONs and tasks to learn without a heavy re-engineering process. VEEGO aims, in the future, at improving students' motor skills, acquire good postural practices and prevent long-term musculoskeletal disorders (MSDs).

1 INTRODUCTION

Dental training begins with a preclinical phase where students train on conventional simulators (fig. 1), also known as "phantom", consisting of a mannequin head equipped with an artificial jaw and teeth (Bandiaky et al., 2024). The main pedagogical objective is to learn care protocols made of steps to treat the tooth. The academic evaluation relies in particular on the respect of the ordered sequence of steps, and the results obtained on the tooth (Bandiaky et al., 2024).

The technical gesture, defined in this context as the sets of body motions and postures representing the motor aspects of a gesture (Vadcard, 2022), are hard to perceive without being close to students. The gestures are also the key factors to enhance performances, respect the patient space integrity and prevent Musculoskeletal Disorders (MSDs). Furthermore, practical sessions typically involve numerous students making impossible, for the teacher, to assist more than one student at the same time (Hefied et al., 2024).

In recent years, motion capture (mocap) systems and Virtual Learning Environments (VLE) has grown extensively given their abilities to simulate learning situations in safety (Michalski et al., 2019), enhance autonomy of learners (Matsuura et al., 2013) with multimodal feedbacks guiding them in real-time (Sigrist et al., 2013). Among the visual feedbacks, 3D anthropomorphic virtual avatar has been used to demonstrate the gesture to learn and analyse motor performances(Le Naour et al., 2019; Westergaard et al., 2024; Manghisi et al., 2022; Shibata and Tanaka, 2023).

There are also virtual and haptic industrial simulators for dental training, such as HRV Virteasy Dental™, NISSIN Simodont™, the SIMtoCARE Dente system™and the UNIDRAW™dental simulator[1]. These simulators incorporate force-feedback arms to reproduce the physical contact between instruments (*e.g.* turbine, contra-angle) and virtual teeth, and sometimes include a phantom head without the physical jaw. However, those simulators only

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[1] Virteasy website, Simodont website, SIMtoCARE website, UNIDRAW website

track motions of the haptic arms and therefore cannot fully evaluate the motor aspects.

Despite their benefits, most current VLEs dedicated to gesture learning fail in incorporating tasks they are not designed for, as well as new observation needs without a significant reengineering process (Oagaz et al., 2022; Senecal et al., 2020). An Observation Need (ON) is defined, in this work, as the expression by the teachers of a set of geometric, kinematic and/or dynamic features of the gesture using their domain-related vocabulary. This set must be respected or avoided by learners. Using supervised Machine Learning (ML) can counterbalance this issue nevertheless, such past works often lack relevant pedagogical feedbacks to correct the learner's gestures (Sallaberry et al., 2024; García-de Villa, 2022). In addition, most current VLEs rarely consider an open-ended multi-expert protocol to get the motor expertise of several teachers. The open-ended term indicates, in this paper, that each expert must be consulted individually and can freely express and demonstrate their motor expertise. This is crucial to identify a broader range of (non)acceptable practices, increase variability in common ones, and strengthen the acceptance of the numerical solution within an expert community. Finally, a recurring limitation of existing studies lies in an insufficient description of the overall pipeline describing the complete processing chain from the data capture to the generated feedbacks in VLE. That goes against the appropriation of the developed system, as well as the assessment of its potential for reengineering when the pedagogical situation evolves.

Consequently, the following **research question** is formalised: *How to design an appropriate VLE pipeline for the evaluation of dental gestures, that integrates the motor expertise of several teachers and their domain-specific vocabulary, while minimising reengineering as observation needs or the task to learn evolve?* This paper presents VEEGO, a Virtual Environment for the automatic Evaluation of Gestures in Odontology, for the evaluation of dental gesture performed on real phantom, scalable to the evolution of the task to learn or the ONs, while including the variability of expertise coming from several teachers.

The following section studies the state of the art related to the methods for automatically assessing gestures, integrating the motor expertise of teachers, as well as designing VLE and their feedback in the dental context. The overall is studied under the perspective of reengineering and system adaptation. Section 3 presents the VEEGO functionalities and a pipeline for its use, from the building of datasets made of captured demonstrations annotated with the teachers' vocabulary, to the colouring of body parts of animated 3D avatars, indicating to the motor flaw to correct in real-time. Section 4 instantiates this pipeline in an experiment where 5 teachers freely expressed their ONs and demonstrate them in 5 different tasks. The built datasets were analyses to study the data variability and the generated random forest models achieved high F-scores showing the effective generalisation from the teachers' data. Section 5 discusses these results, highlights the benefits and points of improvement of the system, while section 6 lays the foundations for the validation of VEEGO with students. A conclusion ends this paper with the perspectives of this work.

Contributions and scope : This work presents a **new motion-based automatic evaluation** approach based on: (i) two amorphological and configurable descriptors based on joint orientations and directional vectors, enabling users to define geometric features for each ON and, (ii) a multi-output classifier automatically generating random forest models, one per ON, linking descriptor values with each ON. In addition, this approach was experimentally validated in a multi-expert context with 5 different ONs, all being analysed in 5 different care protocols. It demonstrates the effective classification of ONs with high performance on the test set.

A second contribution is the **open-ended protocol for multi-expert data collection** with which each teacher is individually consulted to provide both demonstrations and their ONs used to annotate them. The proposed method based on the free expression of the teachers' needs, flaw extraction, polysemy analysis, video annotation and segmented motion labelling allows having both data consistency and sufficient variability.

The **pipeline and architecture of VEEGO** are the third contribution as they proposed: (i) a motion-based automatic evaluation module coupled to the mocap system and the 3D engine, (ii) consequently, the abilities to generate real-time visual feedbacks in VLE, (iii) while minimising the reengineering process for the reuse in other learning situations

One can note that this paper does not intend to contribute to the artificial intelligence domain as it uses existing ML algorithms. Moreover, it does not address the automatic evaluation of dental procedures already handled by existing industrial simulators (Bandiaky et al., 2024). Finally, the pedagogical impact on learning is not assessed yet at this stage (cf. section 6) given the challenges this study must tackle beforehand. The contributions presented in this paper are a necessary step forward, proposing a generic as-

sessment approach and an architecture integrated into a VLE that enables real-time assessment. Nevertheless, an evaluation of the system's ability to provide relevant feedback has been initiated by comparing it to the feedback that a teacher would give (see section 6). This demonstrates the system's capacity to act as a personalised tutor, whereas a teacher cannot be behind every student at the same time.

2 RELATED WORKS

During **dental gesture learning**, with at least twenty students, teachers face high demand and cannot assist more than one student at the same time. Therefore, teachers often prioritise the procedure and the outcome on the tooth following standards (e.g., Black's Class II cavity) over motor aspects of the gesture leading to repeated errors (Hefied et al., 2024). Official guidelines do not necessarily use the teachers' vocabulary and are based on geometric aspects that are difficult to estimate with the naked eye. For instance, the FDI (World Dental Federation)[2] states,*"the interpupillary line is aligned horizontally not more than 15 to 20 degrees"*. Furthermore, FDI does not address all ONs expressed by teachers, such as adjusting working position around the patient's head to maintain visual access to the operative area. These gaps may hinder the acceptance of a VLE if its design relies on official guidelines without incorporating ground practices. The dental surgical gesture has complex motor aspects, which engages a large part of the body under constraints regarding the posture, the positioning around the patient's head, and the control of tools (Hefied et al., 2024). This complexity is amplified by variations in expert practices, often shaped by personal habits (Hefied et al., 2024). For example, instructors may adopt different working position around the patient's head or finger fulcrum for the same task. This necessitates an adaptable VLE integrating several expertises without requiring an excessive reengineering if the needs evolved. That depends in particular on the evaluation method.

To **automatically evaluate gestures**, several methods have been used by digital systems in various domains. Geometric, kinematic and dynamic descriptors can be a first approach (Larboulette and Gibet, 2015). However, experts (except in health and sports domains) do not express their needs by using those mathematical formulas. For example, rather than saying *"the striking angle of the bachi (plectrum) is approximately 42°"*, the ON can be *"a performer should raise their right forearm from the front face of the shamisen and pull down their right forearm"* (Shibata and Tanaka, 2023). Instead of referring to jerk values in m/s^3, teachers might characterise the movement as *"fluid"*. In addition, these descriptors are often: (i) non-trivial to implement, when dealing for instance with complex postural angles (Bhatia et al., 2020)) and defining their acceptability values and rules ((Cóias et al., 2022)) and, (ii) time-consuming as a significant reengineering is needed in case of evolution of ONs.

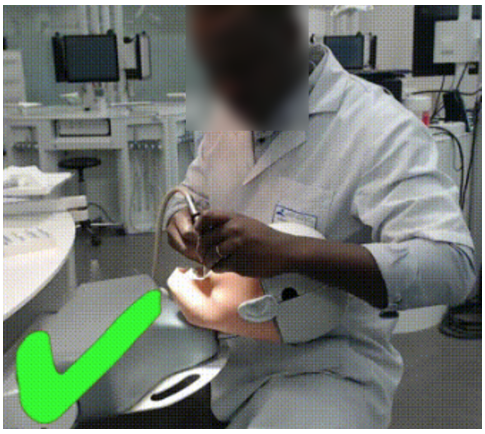


Figure 1: An example of an expected posture

Standardised scores such as RULA (Rapid Upper Limb Assessment) and REBA (Rapid Entire Body Assessment), rely on domain-specific rules and geometric considerations to give an interpretable score. For instance, RULA evaluates postural risks during the task with a score ranging from 1 (low risk) to 7 (high risk). Widely used in industrial ergonomics (Manghisi et al., 2022), RULA has also been employed in dental practices to assess postural risks (Maurer-Grubinger et al., 2021). These standardised scores fail to integrate to evolving ONs as the set of rules is not editable. In addition, the score cannot be used to guide the learners if it is uncorrelated to the domain vocabulary. This last point also concerned the similarity scores, quantifying the proximity of a trainee's gesture with an expert's. DTW (Dynamic time Warping), for instance, can be used to compare the shape of time series representing motion trajectories between learners and teachers in tai chi or rugby (Liu et al., 2020; Le Naour et al., 2019). While this approach allows experts to provide new demonstrations with few or no recoding process, it still requires the manual definition of acceptability thresholds.

Another evaluation method consists in training ML models with motion data annotated by experts. Once trained, the model is expected to generalise and assign appropriate labels to unseen data. (Li et al.,

[2]FDI website

2024) and (Sallaberry et al., 2024) applied ML models to data related to tool trajectories in the respective context of neurosurgical tumour resection and dental anaesthesia. They automatically classified participants by skill level, respectively junior, senior and post-resident, and novice vs. expert. In physical therapy, (García-de Villa, 2022) trained ML models to classify the type of rehabilitation exercises and its correctness, *i.e.* correct vs. incorrect. By providing new motion data as references without recoding, and finding acceptability thresholds, supervised ML methods can address the reengineering problem when ONs or the task evolve. However, inferring an overall appreciation is not sufficient to explain the gesture characteristics that must be improved. In addition, the data collection protocol must be described with details to carefully built the dataset, and a variety of practices must also be included that is often not the case.

Four main strategies for **integrating the motor expertise of several experts** can be found in past works. Editing tools often available in dental haptic simulators such as in Virteasy Dental™, Simodont™, and SIMtoCARE™,[1] enable the creation and modification of exercises and the import of patient data (*e.g.*, STL scans). This integration remains limited to a procedural dimension, with no or poor consideration of the motor aspects.

Consensus protocols can be used, for instance, to select performance metrics on which experts agree on. (Westergaard et al., 2024) conducted such a survey with 27 colorectal surgeons, who rated those metrics on a Likert scale. (Senecal et al., 2019) used semi-structured interviews with salsa dance experts to iteratively refine movement criteria. Such approaches are limited to predefined ONs influencing the expert contributions. Furthermore, defining acceptability values for descriptors remains a problem when not provided by experts.

Another strategy is to integrate demonstrations of several experts. (Morel et al., 2017) proposed to build a nominal motion aggregating several expert gestures into a single reference movement, by using in particular a DTW-based computation. Such a fusion generated an artificial motion questioning its biomechanical consistency.

Supervised classification of gestures into expertise levels can also involves a significant number of participants including experts. It was used in neurosurgery (Li et al., 2024), neonatal intubation (Zhao et al., 2020), dentistry (Sallaberry et al., 2024), *etc.* Gesture recognition through ML implied several experts as annotators in breast palpation for instance (Ourahmoune et al., 2024). However, these kinds of systems only identify the type of gesture without assessing them. In most of the presented integration strategies, the experts are not free to demonstrate and formulate their expertise they want. Outputs allowing learners to improve their performances are absent.

A description of the **pipeline** including the data collection protocol, and the **system architecture** is crucial for the reproducibility and reuse of past contributions. A clear description of the overall processing chain is often omitted in the literature, especially in the context of dental gesture. This hinders the assessment of systems' potential in particular for reengineering. Past works often mentioned the mocap technologies such as: (i) marker-based optical systems (*e.g.*, Vicon™and OptiTrack™), (ii) markerless optical systems (*e.g.*, MediaPipe (Rodrigues et al., 2022)), (iii) inertial systems (*e.g.*, Xsens (Maurer-Grubinger et al., 2021)) and, (iv) haptic devices (*e.g.* Geomatic Touch (Sallaberry et al., 2024)) and VR headsets with embedded tracking (Senecal et al., 2020). Each category offers distinct advantages (*e.g.*, accuracy, portability, cost) and drawbacks (*e.g.*, occlusion, drift, intrusiveness), which influence both data quality. Another often overlooked step is the extraction, from the expert's raw discourse, of the main criteria of their evaluation. In terms of architecture, some current studies present evaluation modules decoupled from the VLE, which do not make possible real-time feedbacks (Sallaberry et al., 2024; Senecal et al., 2020).

Finally, another key point lies in the **design of VLE feedbacks**. Depending one the modality, feedback can effectively convey or enhance, for instance: (i) (visual) spatial information (Le Naour et al., 2019), (ii) (audio) a perception of temporal aspects(Senecal et al., 2020) and, (iii) (haptic) guidance through touch or force (Westergaard et al., 2024) and vibration (Shibata and Tanaka, 2023). Multimodal feedback can leverage their complementary, however, an overloading information effect can also occur (Sigrist et al., 2013). In the context of dental surgery, the extreme accuracy required by the hand and fingers holding the tools, cannot be influenced by a vibration haptic system. The audio feedback will face a noisy environment caused by the contra-angle and turbine noise. Body-part colour coding on 3D avatars has been effectively used to indicate performance errors or correctness, for instance, with green or red to reflect its proximity to the expert motion (Liu et al., 2020). Another effective method consists in displaying the expert's avatar alongside the learner's, for spatial comparison (Le Naour et al., 2019).

Considering the aforementioned points of improvement of past works, the next section presents a full pipeline and the VEEGO system dedicated to the dental gesture evaluation in real-time. The overall

processing chain integrates the collection of the motor expertise and ONs of several teachers. It is also scalable if the ONs or the task to learn evolve, and provide visual feedbacks, in VLE, guiding the learners on the gesture features to improve.

3 PIPELINE & SYSTEM DESIGN

What is VEEGO? VEEGO is a VLE, projected on a screen, where the dental learners' gestures are evaluated in terms of motor aspects in real-time (fig. 2). VEEGO aims at capturing students' motions while working on a real phantom in preclinical situations. Meanwhile, the VLE displays the learner's 3D avatar coloured according to the detected ONs also represented by texts. The ONs are detected by using binary random forest models, one per ON, trained beforehand on teachers' demonstrations. The demonstrations are recorded, segmented and labelled using a proprietary mocap software and VEEGO. Finally, once the task is finished, a report is available to see the most prevailing ONs. To achieve this, VEEGO is structured around two operating modes: training mode and evaluation mode.

In **VEEGO training mode** (fig. 2 (a)), the user segments each teachers' demonstration streamed from a proprietary mocap software (*e.g.* QTM™, Axis Studio™), that is previously used to capture teachers' motions. An ON label is associated to each motion segment. The *Train button* allows generating the dataset made of descriptor values computed from motion segments, and training or enriching the random forest model associated with the selected label.

Once all ML models are trained, the **VEEGO evaluation mode** can be used to assess the learner's gestures in real-time, thanks to the streaming of the mocap priority software. Feedback is provided through a colour applied to the 3D avatar's body parts according to the detected flaws. Additional texts show, in real-time, each ON label in red when the ON is detected, and in green otherwise (fig. 2 (d) top). After the practical work, a textual statistical report summarises the occurrence percentage according to the task duration of each detected flaw (fig. 3 top). Flaws exceeding a predefined occurrence threshold of 20% can then be reviewed with the replay function: the implied body parts are highlighted in red on the student's avatar and green on the teacher's. Both avatars can be visually compared, the teacher's avatar being animated with a correct demonstration.

3.1 Pipeline

The proposed pipeline describes the complete data processing chain, from the protocol for multi-expert data collection to the feedbacks provided to the learner in VLE.

In the **open-ended protocol for multi-expert data collection** (fig. 4), each teacher is first asked to formalise non-acceptable practices, which are transcribed as notes. The teacher's gestures are then recorded while performing demonstrations of these flaws as well as some correct demonstrations on a real phantom.

Motion data are represented as a sequence of frames, each defined by a frame time and a joint tree. Each joint is described by its 3D position and orientation. The *motion filtering* step corrects the raw motion data from noise, inconsistencies or gap by applying filters (*e.g.*, QTM's polynomial and relational filters[3]). During the *label filtering* step, each teacher's description is formulated into one or several ON labels each representing a flaw. The subsequent *polysemy identification* step consists in aggregating semantically similar ON labels. Videos recorded during the mocap process are manually annotated with the labels, that also allows the identification of the beginning and ending time steps of each flaw. Afterward, to get several possible motor variations for each ON, the *label variability validation* step excludes labels that are not shared by a predefined minimum number of teachers and/or demonstrations, to get a final list of N aggregated and validated labels.

Then, by using the proprietary software of the mocap system, each recorded and filtered motion (clean motions on fig. 4), related to the N validated labels, is streamed into VEEGO, which applies a *calibration*. This procedure expresses position and orientation data in a coordinate system aligned with the real phantom placement, whose position and orientation were also captured (fig. 5 (c)). This ensures the invariance of motion data as the mocap system setup can generate a new coordinate system at each installation.

The *motion segmentation and labeling* step extracts the motion segments, each corresponding to a single state of one validated ON label *i.e.* its observation (KO) or not (OK). The user specifies a user ID, a trial number, one of the validate labels, its state and the motion segment duration (fig. 2 (a)). The beginning and ending time steps of the flaw are set in the mocap software. By pressing the "GO" button, VEEGO extracts the motion segment from the demonstration and labelled it. This step also elimi-

[3]Qualisys website

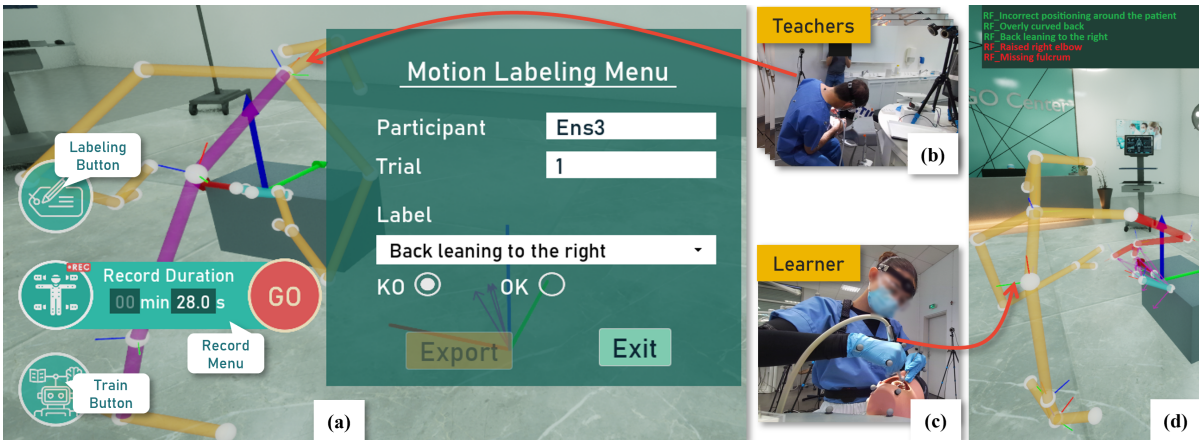


Figure 2: VEEGO overview: (a) training mode, (b) teachers' demonstrations, (c) learner on phantom, (d) real-time feedback ("Raised Right Elbow", "Missing Fulcrum").

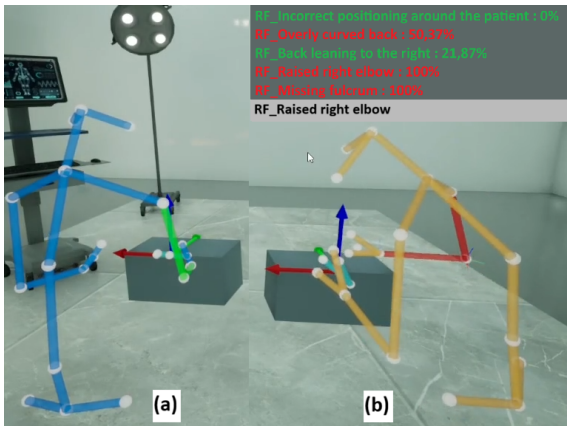


Figure 3: VEEGO automatically replays the teacher's correct demonstration (a) and the student's performance (b)

nates idle motion parts (*e.g.*, taking a tool, stretching, pause, *etc.*).

During the **descriptor configurations** step (fig. 6 (b)), each ON will be characterised by geometric features computed from two amorphological descriptors: *Directional Vectors (DV)* and *orientation* (expressed with a quaternion). Both are implemented in VEEGO as functions whose parameters are defined by the user in a file, one per ON label, whose name is placed at the file beginning. For an orientation, the parameters are: (i) the list of joint names and, (ii) the reference frame from which the quaternion is computed (local vs global). For a DVs, the parameters are: (i) the start and the end joint names (in case of multiple start and end names, each start joint will be combined with each end joint, leading to the generation of multiple directional vectors), (ii) their computation relative to a reference joint if necessary and, (iii) the normalisation of the vector or not.

The **dataset building principles** rely on the creation of N datasets, one per ON label, made each of geometric descriptor values. The creation of a dataset is done as follows: (i) given a motion segment associated to an ON label state, the corresponding set of geometric descriptors values are computed at each frame, (ii) each set and the state are added into the dataset, (iii) the dataset is parsed once using a sliding window of 3 frames to delete consecutive values exceeding 0.0005 defined as a similarity threshold and, (iv) the three previous steps are repeated for all the motion segments associated to the same label.

The chosen **ML model to train** is Random Forest (RF), as it can handle heterogeneous input spaces (*i.e.* Euclidean space, geodesic space, *etc.*) without requiring a specific distance metric. In addition, this model is also well known to work with limited amount of data, as it is unrealistic to expect a large number of demonstrations from teachers(Joudieh et al., 2023). Based on the identified research objectives, the following hypothesis is formalised. **H1** : RF can capture the variability of the same motor-based ONs, *i.e.*, for one need, it can converge on labelled data coming from different experts' demonstrations from which the same set of descriptors is computed. The *RF Generation and Training* step consists of triggering, *via* VEEGO, the generation of the N datasets that are then saved in files. Each dataset is split into two subsets: 80% for training and 20% for testing. The splitting method respectively considers 80% and 20% of each set of descriptor values computed from each motion segment. A 5-fold cross-validation is applied to the training set to define the number of trees and the best-performing model is selected for inference. In the end, N binary RF models, one per ON label, are serialised and saved in files.

The **learner evaluation** relies on a real-time capture of the learner's motions (fig. 6 (a)). The learner's

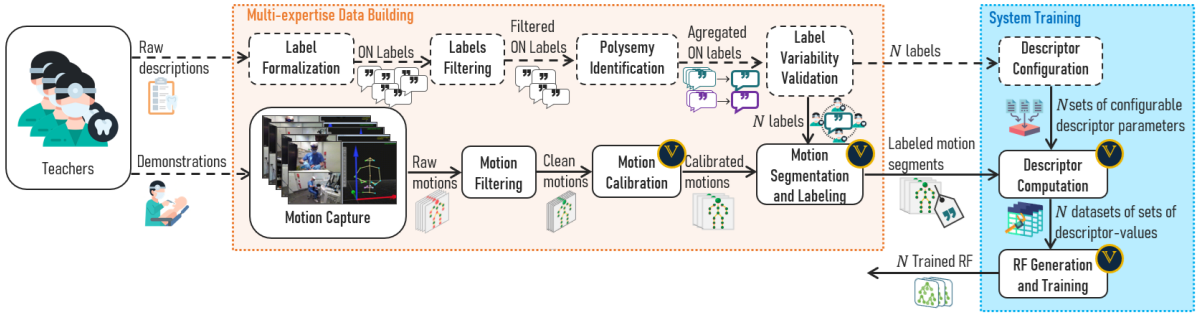
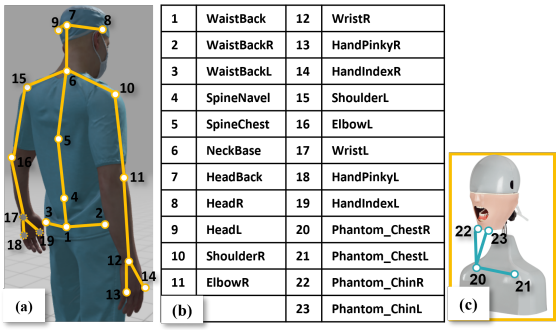


Figure 4: Training phase of the pipeline: dashed blocks and arrows respectively present manually executed steps and provided data by users; blocks marked with "V" are executed within VEEGO.



1	WaistBack	12	WristR
2	WaistBackR	13	HandPinkyR
3	WaistBackL	14	HandIndexR
4	SpineNavel	15	ShoulderL
5	SpineChest	16	ElbowL
6	NeckBase	17	WristL
7	HeadBack	18	HandPinkyL
8	HeadR	19	HandIndexL
9	HeadL	20	Phantom_ChestR
10	ShoulderR	21	Phantom_ChestL
11	ElbowR	22	Phantom_ChinR
		23	Phantom_ChinL

Figure 5: (a) Participant joint tree; (b) Joint names; (c) Phantom joint tree

data undergoes a *calibration* step aligning each frame in a similar way to the *motion calibration* step. The *descriptor computation* step generates the N sets of descriptor values, one per ON label, from the current frame. Each of the N sets is subsequently provided as input to the corresponding trained RF model. Each RF model outputs a binary decision *i.e.* the ON is observed (KO) or not (OK). The collection of these N predictions is used by the visual feedback in VEEGO in real-time.

3.2 Architecture

The Unreal Engine 5 (UE5) through blueprints and C++ coding, as well as Python were used to implement VEEGO (fig. 7). A Python server communicates with the UE5 part using WebSocket. Thanks to the UE Livelink plugin, UE5 can handle several mocap hardwares and softwares (*i.e.* Qualysis™infrared cameras with QTM™, Perception Neuron™with Axis studio™software and Azure Kinect™with the dedicated Microsoft SDK). The UE5 part is in charge of: (i) the creation of the labelled motion segments saved in BVH files, (ii) after loading the descriptor parameters from JSON files, the generation of the set of descriptor values, being saved in a dataset stored in CSV files for the RF training, or directly sent to the Python server for the real-time evaluation, (iii) triggering the RF training or the motion evaluation from RFs and, (iv) the user interface and the display of the VLE. The Python part is in charge of: (i) the generation and training of RFs models, using scikit-learn, with the sets of descriptor values loaded from the CSV files, the RFs being serialised in JOBLIB files and, (ii) after loading the RFs from the JOBLIB files, the real-time motion evaluation from the set of descriptor values sent by the UE5 part through WebSocket.

4 EXPERTISE INTEGRATION

This section presents the experimental setup and protocol used to collect the observation needs and motion data, as well as the application of the proposed pipeline to train and test the VEEGO system. The experimentation was conducted in a preclinical simulation classroom at the UFR of Odontology, Nantes, France, with a capture laptop (Intel® Xeon® W-11955M CPU, NVIDIA RTX A3000 GPU, 32 GB RAM), 8 Qualisys Miqus M3 ™cameras, and 2 video cameras surrounding the phantom head (fig. 2(b-c)). The marker set is shown in fig. 5.

4.1 Experimental Protocol

Each teacher execute a dental procedure of their choice on a real phantom. They are asked to prepare their procedure, the target tooth, and the required instruments. The mocap system is calibrated using the Qualisys QTM™software and the teacher is equipped with markers. An ID is given and their dominant arm is noted. Before the work starts, the chosen procedure is also noted. Then the teacher is asked to: (i) describe a motor or postural aspects considered unacceptable, which is transcribed and, (ii) perform a demonstration of this flaw under motion capture. These two steps are repeated for each flaw the teacher

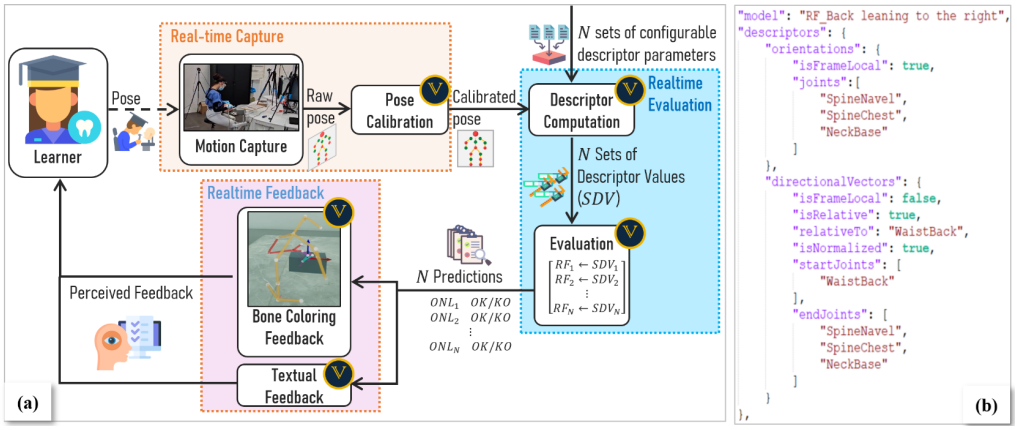


Figure 6: (a) Evaluation phase of the pipeline: dashed blocks and arrows respectively present manually executed steps and provided data by users; blocks marked with "V" are executed within VEEGO; (b) Example of a descriptor configuration file

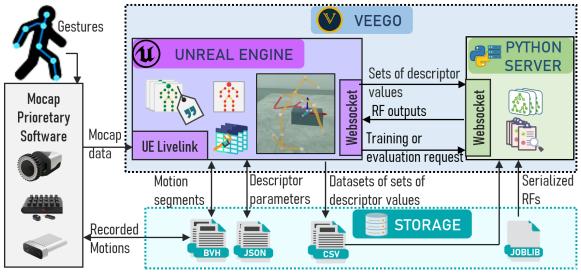


Figure 7: VEEGO architecture

wishes to provide. The protocol ends with the record of two flawless demonstrations of the teacher. The description noted at step (i) is named "original description".

4.2 Training Pipeline Application

A total of 5 right-handed teachers, each named TX (X: the teacher number), performed their procedure *i.e.*: *preparation of an occlusal carious cavity on tooth 85* (T1), *peripheral crown preparation on tooth 26* (T2), *occlusal cavity on tooth 26* (T3), *zirconia crown preparation on tooth 16* (T4) and *pulpotomy with cavity preparation on tooth 85* (T5). **29 original descriptions of unacceptable gestures**, divided up in 4, 5, 5, 10 and 5 respectively from T1 to T5, **and 10 flawless demonstrations were recorded**.

Each original description can be decomposed in several labels when it expresses several flaws. For instance, T1 described *"variation in elbow inclination: elbow moved away from the trunk or raised to the right side at different levels"*, which was decomposed into *"Elbow moved away from the trunk"* and *"Elbow raised on the right side"*. Two annotators, with experience in dental gesture, extracted those labels from the original description and then performed the polysemy analysis aggregating common ONs expressed in different ways. For example, T2 described *"Working position abnormally varying between 12h and 2h"* and T3 said *"Practitioner incorrectly positioned at 1h"* that can be aggregated into the label *"Incorrect positioning around the patient"*.

By consensus, **14 different aggregated labels were inventoried**: *"Incorrect positioning around the patient", "Overly curved back", "Back leaning to the right", "Raised right elbow", "Missing fulcrum","Practitioner's head too close to the patient's", "Seat too high", "Patient's head in hyperextension","Practitioner's hand on the patient's head","Bad handling of the tool","Practitioner's arms on the patient's chest","Excessive tilt of the patient's head","Patient's head too high","Elbows bent"*. The "OK" label was also added for the flawless demonstrations.

Afterward, the experts separately annotate the demonstration videos with those labels. **50 and 55 aggregated labels were respectively observed by each annotator**. One can note that it is possible to introduce an aggregated flaw label detected in other videos than the video used to specifically record the flaw. The Cohen's κ coefficient was computed and is 0.86 meaning a "quite perfect agreement" among the annotators (Landis and Koch, 1977).

To improve the flaw variability, only the aggregated labels coming at least from "3 teachers" or "2 different teachers with at least 3 demonstrations" were retained. In the end, the following **5 aggregated and validated labels were retained** (cf. tab. 1, third column for the number of teachers, fourth column for the number of motion segments): *"Incorrect positioning around the patient", "Overly curved back", "Back leaning to the right", "Raised right elbow", "Missing fulcrum"*.

Table 1: Aggregated labels: date regarding the involved teachers, motion segments, temporal aspects and demonstration variations. Time is in minutes. "Seg." stands for motion segments.

Label		Number of	Number of seg.		Seg. duration (minutes)			% of the retained	F1 score	
No.	Name	teachers	KO	OK	OK+KO	KO	KO (mean)	frames	Train	Test
1	Incorrect positioning	3	4	3	03:12	01:42	00:25 ± 00:08	11.83%	1.0	1.0
2	Overly curved back	4	5	3	03:57	02:27	00:29 ± 00:01	44.89%	1.0	1.0
3	Back leaning to the right	2	3	4	02:38	00:59	00:19 ± 00:16	49.72%	1.0	1.0
4	Raised right elbow	5	6	3	04:19	02:49	00:28 ± 00:02	54.4%	1.0	1.0
5	Missing fulcrum	3	3	3	02:55	01:25	00:28 ± 00:02	96.96%	1.0	1.0

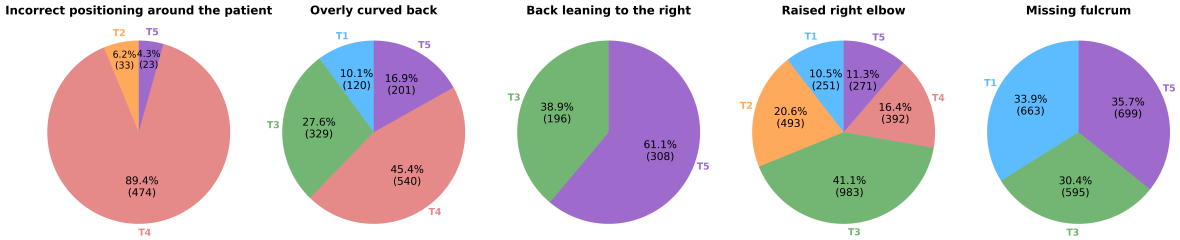


Figure 8: % of number of frames in the training dataset per teacher per model

Each raw motion data related to each motion segment were cleaned using Qualisys QTM™ software (cf. section 3.1). **The 21 motion segments of the 5 validated labels were extracted** by VEEGO once the start and end frames were manually defined by consensus of the annotators.

The **descriptor configurations** were set for each five label as follows (number in parentheses are the joint number seen in fig. 5):
- *"Incorrect positioning around the patient"*: local orientation of the pelvis (1) and Normalized DVs (NDV) connecting it to the thoracic area (20, 21);
- *"Overly curved back"* and *"Back leaning to the right"*: local orientations of the lumbar (4), dorsal (5), and cervical (6) joints, together with their relative positions to the pelvis (1) expressed as NDVs;
- *"Raised right elbow"*: local orientations of the forearm joints (11, 12) and an NDV representing the position of the elbow (11) relative to the shoulder (10);
- *"Missing fulcrum"*: local orientations of the hand joints (12, 13, 14) and the phantom chin (22, 23), as well as DVs linking the hand to all phantom points;

4.3 Results

Tab. 1 reports the statistics of the built datasets.

For each aggregated label, the mean time of a segmented motion of flaw is between 20 and 29 seconds. The total duration of all motion segments *i.e.* flaw (KO) and flawless (OK) goes from 2 minutes and 38 seconds to 4 minutes and 19 seconds. The total duration representing a label (KO), goes from 1 minute and 25 seconds, to 2 minutes and 49 seconds. Various % of the retained frames compared to the total duration can be noted after datasets building. The second, third and fourth labels used between 45 and 54% of the segmented motion data, meaning that only half of the data offer variations among the teachers. For the fifth label, the demonstrations seem to offer a significant variability as 97% were used. On the contrary, only 12% of the data were used for the first label that could mean very close demonstrations.

Fig. 8 shows the contribution of each teacher, in terms of % of used frame numbers, to the building of the training datasets. For *"Missing fulcrum"* and *"Back leaning to the right"*, teachers contribute in a quite equal way. This may increase the variation for these labels. The contribution of T3 is, at least, twice as much as other teachers for *"Raised right elbow"*. The same thing can be said for the *"Overly curved back"* with T4. In both cases, the proportion of each teacher is significant to contribute to the flaw variability. On the contrary, T4 is the main contributor for the *"Incorrect positioning around the patient"* that questions the variability and is consistent with the small amount of retained data.

For each label, an RF model was trained with their own sets of descriptors computed from their own set of motion segments. 100 trees were set after the 5-fold cross validation process. All models achieved a perfect F1 score on the training and test sets (cf. tab. 1, last two columns). Each ON was perfectly recognised and the generalisation process is hopeful, while the integration of multi-expertise is validated at least for the last four labels.

5 DISCUSSION

The proposed pipeline based on the VEEGO system succeeded in generalising the detection of 5 ONs from teachers' data. Except for *"Incorrect positioning around the patient"*, the 4 other models include demonstrations coming from several experts. The data collection protocol is open-ended as each teacher is consulted individually and remained free to express and demonstrate their own motor expertise. It enables, together with VEEGO, the construction of evaluation models integrating the variations in teachers' practices for the ONs they share. Therefore, H1 is validated in the specific context, the expressed observation needs and performed demonstrations. This can improve the system acceptance as it can capture better the richness of experts' skills.

This work was done without coding ad-hoc descriptors or acceptability rules for each ON, thanks to (i) the amorphological and configurable descriptor approach, allowing users to focus on the specification of the implied body parts, the posture configuration and spatial relations between the user and the phantom, (ii) the automatic generation and training of a RF per label and, (iii) the binary multi-output strategy that makes the system scalable. Indeed, configuring the descriptors in the JSON file for a new ON and give labelled motion segments is sufficient to enrich the evaluation capabilities. In addition, VEEGO can also integrate additional training to reinforce existing labels without altering the existing label space or dataset structure.

The supervised ML approach used in VEEGO is not intended to classify performance levels (*e.g.*, expert vs. novice), produce an overall task appreciation (*e.g.*, correct/wrong) or recognise actions without significant pedagogical feedbacks. Instead, it links low-level descriptors to the teachers' ONs, to indicate which specific aspects of the gesture must be corrected and how to correct it. Indeed, VEEGO aims at detecting observed flaws, displayed in real-time, through the coloured learner's avatar in VLE. Furthermore, as an appraisal, the teacher can choose one of the detected flaws, to visually compare the learner's performance with the teacher's thanks to their 3D avatars in VLE (cf. footnote for video demonstrations[4]). This work hopes to show its ability to build models matching the observation needs of teachers while giving comprehensive visual feedbacks to learners following an observation-imitation process. However, VEEGO must be: (a) validated in work sessions implying students and (b), studied from the perspective of its pedagogical benefits (cf. section 6).

Several points of improvement must be considered. For this work, infrared cameras and a marker set provided high-quality data. This first and expensive strategy ensured that potential errors, during VEEGO development, do not come from the data quality. The pipeline being validated, the robustness in building flaw models will be tested with other kinds of mocap hardwares such as inertial unit-based ones or optical-markerless ones.

The open-ended nature of teachers' contributions can lead to naturally unbalanced datasets. For *"Incorrect positioning around the patient"*, teacher T4 is the main contributor that can affect the system robustness. However, this must be tested with student's gestures, to confirm/infirm its generalisation capacity. In addition, significant annotation efforts are required to interpret heterogeneous discourses, align semantically descriptions, and segment motion data. Furthermore, VEEGO was trained with data coming from volunteer right-handed teachers. Left-handed teachers must be found to enrich the datasets and adapt VEEGO to left-handed students.

Defining effective descriptor configurations still requires some knowledge in biomechanics. A lead could be to implement every combination of parameters for orientations and DVs for every ON. As RFs converge by testing a subset of its input, the results could automatically be analysed to find the best descriptors for each ON. In addition, kinematic-based ONs (*e.g.* based on velocity, acceleration, jerk, *etc.*) will require the extension of this approach with time-based parameters and the study of RF models with time series.

6 PEDAGOGICAL CHALLENGES

In dental surgery, students learn care protocols on any of the 48 teeth belonging to the four quadrants of the mouth (2 on the mandible and 2 on the maxilla) (Kamal et al., 2020). When working on the maxilla, the recommendation is to work in indirect vision by using a mouth mirror (Bud et al., 2021), to avoid incorrect postures (*"Incorrect positioning around the patient","Overly curved back", "Back leaning to the right"*). Nonetheless, this requires to learn gestures observed in a mirror. In addition, the recommendations in terms of fulcrum and positioning also depend on the tooth position in the quadrants. Overall, in the first years of study, students often adopt incorrect body configurations that can lead to MSD (Kamal et al., 2020).

VEEGO could counterbalance this phenomenon

[4]Public data repository

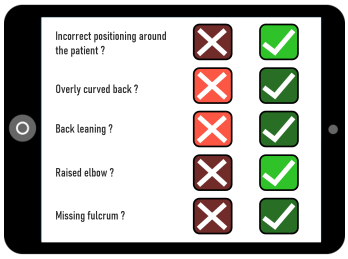


Figure 9: Tablet application to report each of the 5 flaws in real-time

with its real-time feedbacks. However, does VEEGO able to correctly detect flaws of students from the built models in this work ? A method consists in building an experimental protocol with a comparison between VEEGO outputs and teachers' evaluation. This is not a trivial task, as the evaluation of VEEGO is executed on motion frames with a frequency that the teacher cannot handle. In addition, not matter the evaluation comes from (VEEGO or the teacher), the detection of the flaw at a given time could be wrong. Consequently, to compare each evaluation, the following points must be considered: (i) the time of the detected flaw (ii), the time period where a flaw starts and ends and (iii), the ordered sequences of time periods where flaws are detected. Using the time periods and their sequencing will offer a basis to evaluate the robustness of VEEGO in comparison with teachers. A simple application on a tablet was developed (cf. fig. 9) where a teacher can point out one of the five flaws in real-time, each evaluation and its time being saved in a CSV file.

An experiment will be conducted where at least 60 students (first two years of study of Nantes Odontology School, France) and 2 teachers will be involved (demonstration videos are accessible in the footnote[4]; cf. fig. 10). Each student will be assigned to one of the fourth quadrants with a tooth chosen by teachers, to work on one of the first scenarios learned during their curriculum *i.e.* a tooth preparation for the crown placement. Their gesture will be captured and evaluated by VEEGO in real-time and each teacher independently with the tablet. The care protocol will be paused at each step (defined by consensus of the two teachers) to make the students see their flaws detected by VEEGO by observing the replay of their 3D avatar. The students will be invited to correct the detected flaw during the next step. The agreement between VEEGO and teachers could be computed thanks to the Cohen's κ coefficient or other Intra-class Correlation Coefficient (ICC) (Landis and Koch, 1977; Koo and Li, 2016).

A pretest was conducted to study the experiment feasibility with the same setup described in section 4.

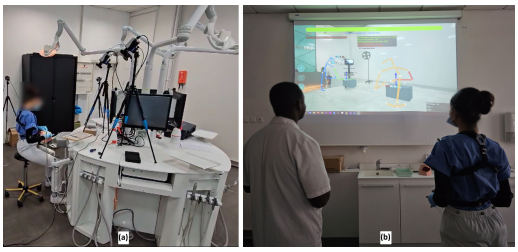


Figure 10: A working student with VEEGO (a); The student and teacher watch the performance replay (b)

The first results considering 3 students show around 30% of marker loss by the mocap Qualisys™system[5] during the execution of the aforementioned task and given the absence of motion post-filtering. After each step, the teacher chooses the main flaws outputted by VEEGO, and the students are invited to watch their 3D avatars replaying the parts implying the flaws. The two experts involved in the pretest sessions do not notice output aberrations during the real-time evaluation, except when a marker loss occurred. The first students' comments were positive such as *"I like the experience and will consider the feedback during future work sessions"*, *"VEEGO allows me to correct a lot of errors regarding ergonomics and my placement around the fantom"* and *"Highlighting the errors allows me to correct them immediately"*. These first results are acceptable in terms of feasibility and give encouraging insights on the pedagogical benefits of VEEGO.

7 CONCLUSION

This work, situated in the context of preclinical dental training, aims at providing teachers and learners with a Virtual Learning Environment (VLE), named VEEGO, capable of a real-time evaluation of the motor aspects of gestures to produce relevant pedagogical feedback. This study presents the entire pipeline, from the data collection related to the teachers' motor expertise to the delivery of feedbacks to learners within a virtual environment.

The main contributions rely on (i) an new evaluation approach structured around two configurable motion descriptors (joint orientations and directional vectors) combined with a binary multi-output classifier (RFs), (ii) an open-ended protocol for the collection of data from several experts, that produces segmented and labelled motion segments by combining expert discourses and demonstrations, these data being used to train the RF models, (iii) an architec-

[5] % of marker loss is directly accessible in real-time in the QTM software

ture integrated into a VLE enabling real-time evaluation and, (iv) a complete reproducible and validated pipeline for its (re)use.

The open-ended nature of the data collection protocol enables the identification of a rich set of (non-)acceptable practices, concretely observed by teachers in preclinical situations, while capturing variations in teachers' demonstrations related to shared Observation Needs (ONs). The evaluation approach demonstrates its ability to integrate 5 different ONs, achieving perfect F1 scores, with minimal reengineering limited to external descriptor configuration. In addition, the proposed architecture facilitates the training of ON-specific models and, during evaluation, supports the co-occurrence of multiple flaws, associated with the simultaneous generation of visual feedbacks.

Future work will focus on validating the system through an experimental study involving students. This study will aim, on the one hand, to compare the system's outputs with the evaluations of teachers and, on the other hand, to gain insights related to the impacts of using VEEGO on learners' performances.

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